


# IK2215: Project assignment

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# Objectives

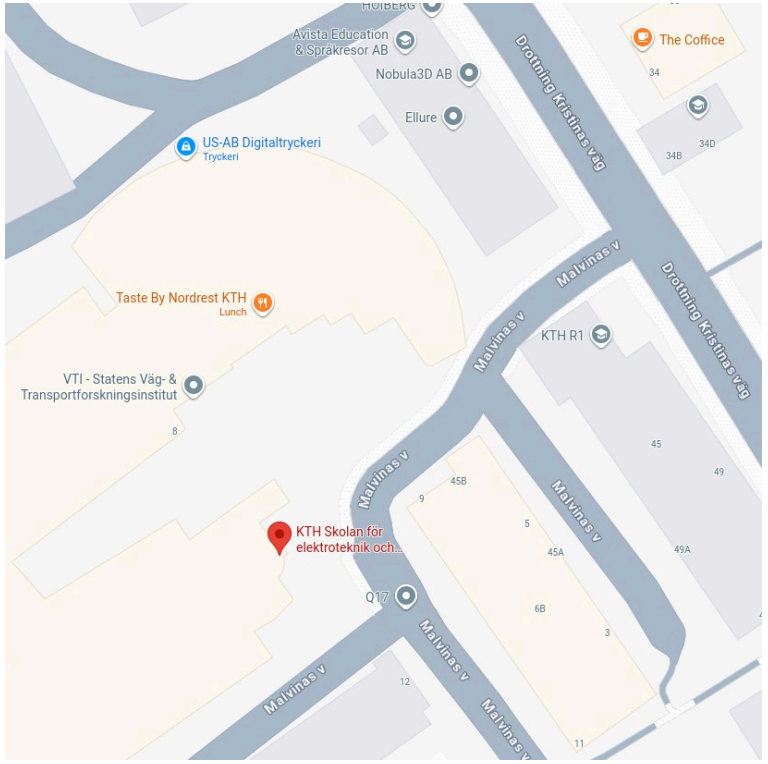
- Hands-on experience in designing, configuring & handling TCP/IP networks, and providing services as an Internet Service Provider (ISP) to end users by primarily using PC hardware and Unix
- Insights into how an underlying network influences deployed services
- Learn to test and troubleshoot configurations of your network and services
- Venue for you to describe and discuss network design and implementation
- Learn to work professionally

You work in pair with an assigned project partner in Project Groups

Contact us ASAP if you are not in a group!

# Overview of the project assignment

- Guideline on the course web (see the Project assignment page)
- Scope of tasks
  - Routing (networking protocols)
  - Internet services
- Deliverables
  - Reports (LaTeX templates in Overleaf)
    - Network design report max 6 pages
    - Peer-review report max 4 pages
  - ISP implementation (on the lab VM)
  - Demonstration
    - Perform individually (Project partners present in different demo sessions!)
    - ISP implementation must pass our verification to be allowed to demonstrate
    - Location: Division of Communication Systems (COS), Malvinas Väg 10, Floor 3.



# Grading overview

- Pass/Fail — You must pass all activities to pass the project module
- Reports (templates on Overleaf) — Total 6 points
  - Network design reports (group) — 2 versions — 2 points each
  - Peer-review report (individual) — 2 points
- ISP implementation — Total 30 points*
- Demonstration (perform individually) — Total 14 points
  - Presentation slide — 4 points
  - Oral presentation — 7 points
  - Q&A — 3 points
- Passing criteria
  - Score at least 1 point in each reports (total 3 points)
  - Score at least 24 points in ISP implementation (Not allow to demo if you fail!)
  - Score at least 10 points in demonstration



* Details of ISP implementation grading is on the next slide

# Details of ISP implementation grading

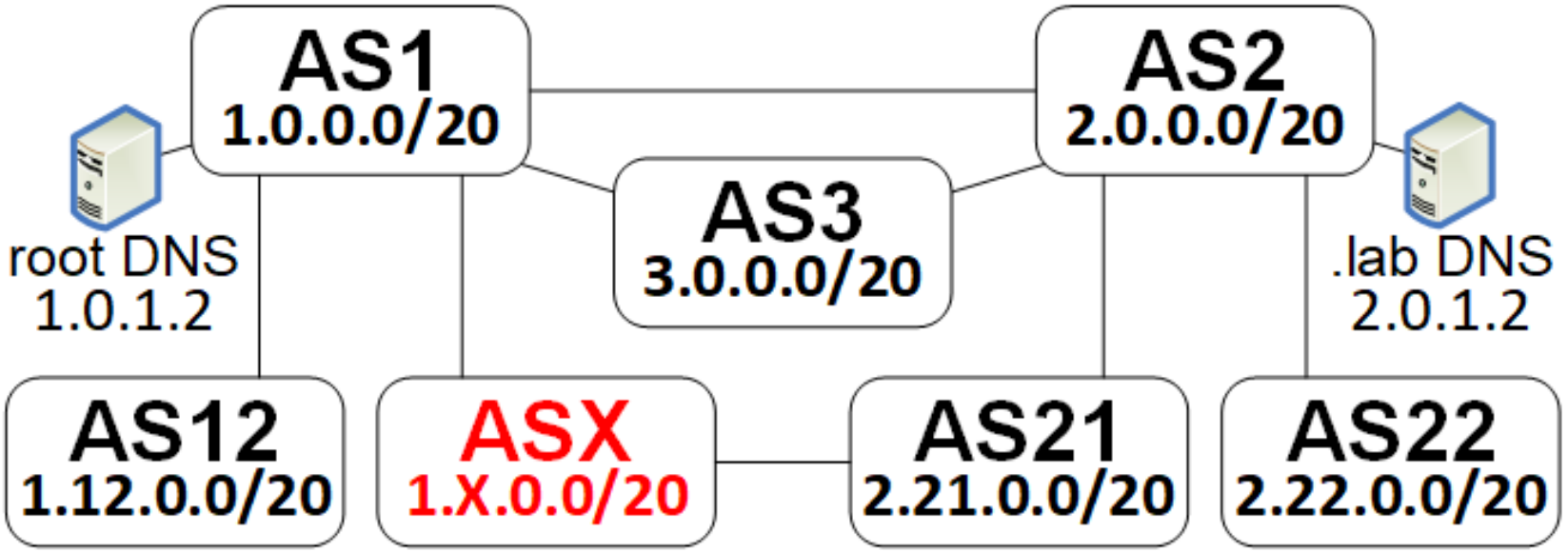
| | |
|---|---|
| basic_configurations | point: 5 of 5 |
| basic_configurations | bonus: 1 |
| dhclient | point: 1 of 1 |
| ping | point: 2 of 2 |
| dns_forward | point: 2 of 2 |
| dns_reverse | bonus: 1 |
| dns_all_interfaces | point: 1 of 1 |
| www | point: 2 of 2 |
| same_path | point: 2 of 2 |
| IGP_disruption | point: 4 of 4 |
| BGP_transit | point: 3 of 3 |
| eBGP_disruption | point: 4 of 4 |
| eBGP_ISP_disruption | point: 2 of 2 |
| iBGP_disruption | point: 2 of 2 |

TOTAL: 32

# Requirements

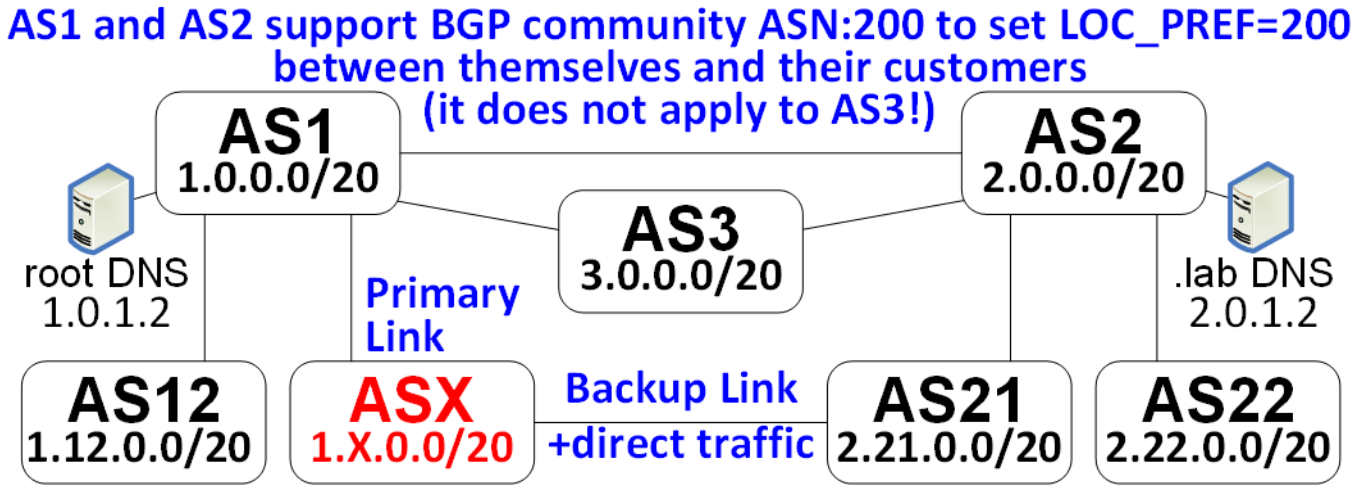
- Routing requirements
  - Intra-domain routing — deterministic paths with redundancy
  - Inter-domain routing — primary and backup links with transit
- Internet service requirements
  - DNS — ns.isp*X*.lab (1.*X*.1.2) — forward lookup (reverse lookup is optional)
  - Web — www.isp*X*.lab — simple text-based web page
  - DHCP — dhcpd.isp*X*.lab — dynamically assign IP for client network
  - *X* is your autonomous system number (ASN), which is 100 + your group number

# Network organization – “our Internet”



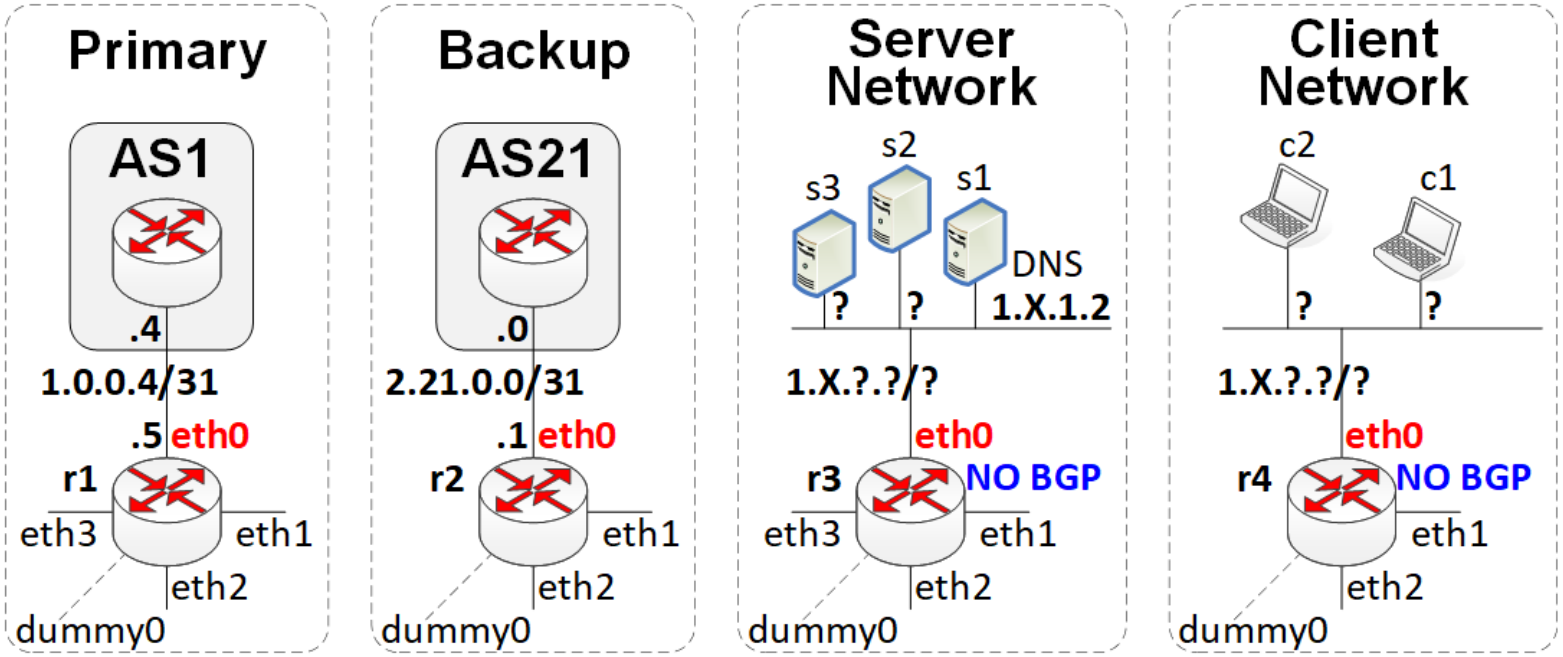

- DNS server: ns.isp*X*.lab
- Web server: www.isp*X*.lab
- DHCP server: dhcpd.isp*X*.lab
- *X* is the ASN: ASN = 100 + <group_number>

# Inter-domain routing policy




- Customers of different providers with a backup link
- Transit service for your neighboring AS (AS21) No transit for other ASes!
- Advertise only 1.*X*.0.0/20 to other ASes and suppress other subnets
- All ASes use default BGP policy
  - AS1 and AS2 support BGP community value of ASN:200 (i.e., 1:200 and 2:200 respectively)
- Your policy must enforce the inter-domain routing policy of all ASes

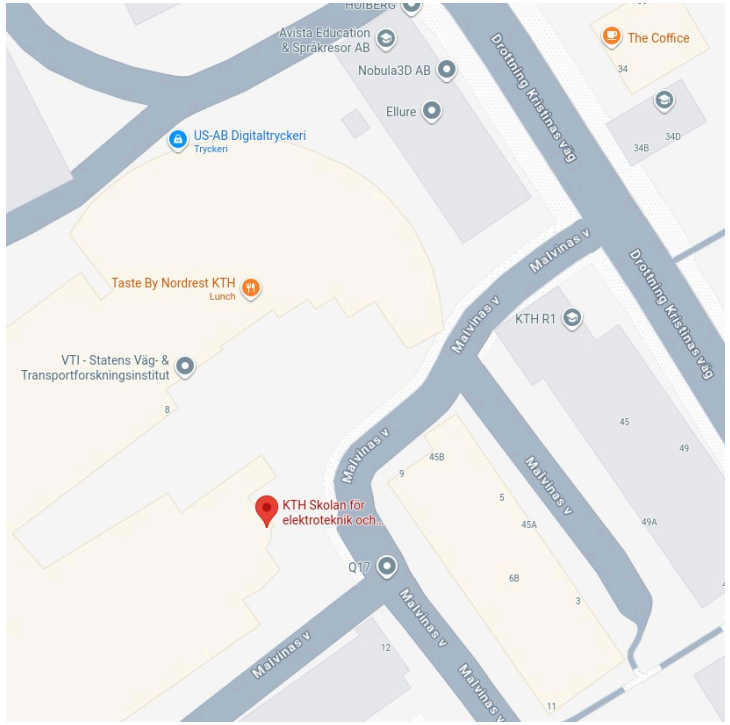
# Resources




- eth0 must connect as shown in the figure.
- Other Ethernet interfaces must be connected as a point-to-point link (i.e., using /31 or /30 subnet)!
- dummy0 for logical interface
- Do not run BGP on r3 and r4! (i.e., they are non-BGP routers)

# Milestones

- 11 Sep: Student pair assignment (group 1 with AS101, 2 with AS102, and so on)
- 11 Sep: Project template files (project_template.tgz on Canvas or project.tgz in lab VM)
- 18 Sep: Lab session: BGP implementation
- 23 Sep: Submit your network design report
- 25 Sep: Submit peer-review report
- 26 Sep: Submit ISP project implementation for a preliminary test
- 29 Sep: Demo session assignment
- 1 Oct: Submit final version of your network design report
- 1 Oct: Submit ISP project implementation
- 1 Oct: Submit one-page slide
- 3-7 Oct: Project demonstration




| Date | Time | Room |
|---|---|---|
| 2025-10-03 | 10:00-12:00 | Bell, COS, Malvinas v 10, Floor 3 |
| 2025-10-06 | 08:00-11:00 | Bell, COS, Malvinas v 10, Floor 3 |
| 2025-10-06 | 13:00-17:00 | Bell, COS, Malvinas v 10, Floor 3 |
| 2025-10-07 | 08:00-12:00 | Bell, COS, Malvinas v 10, Floor 3 |

# What you need to do

- Read the project guideline thoroughly
- Familiarize with Overleaf, i.e., writing a LaTeX document
- Contact your project partner
  - Agree on how to collaborate (e.g., create an Overleaf project for reports)
  - Divide the work equally in the same category
  - Avoid splitting based on services, i.e., routing and internet services
  - Schedule regular meeting to discuss work progress
- Focus on your network design based on information in the guideline
  - Project template (to run with Kathará) include in lab VM and in Canvas
- Testing and troubleshooting are one of your main tasks!
  - Discuss problems with your project partner first before asking others

Report to us if your partner does not contribute their fair share of work!